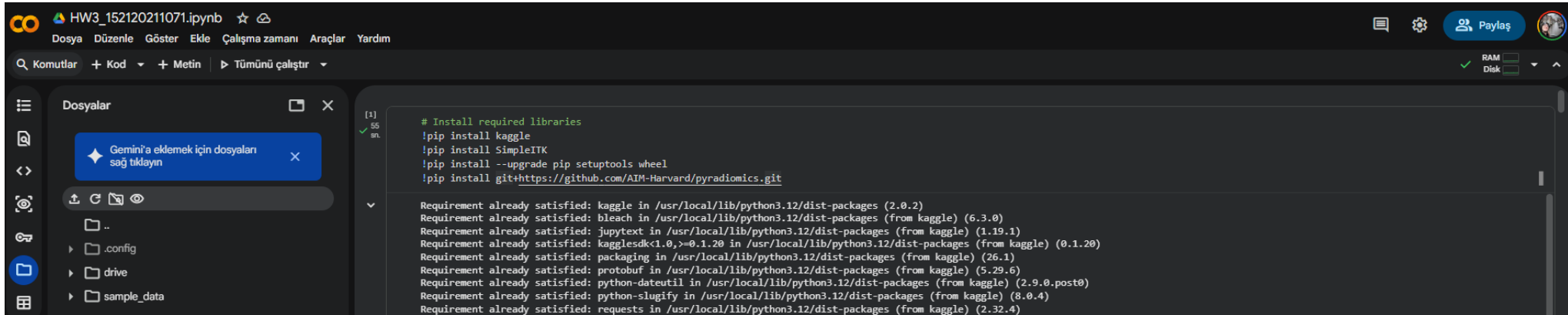


HW3 REPORT



The screenshot shows a Jupyter Notebook interface with a terminal window open. The terminal displays the following commands and output:

```
# Install required libraries
!pip install kaggle
!pip install SimpleITK
!pip install --upgrade pip setuptools wheel
!pip install git+https://github.com/AIM-Harvard/pyradiomics.git
```

The output shows that all requirements are already satisfied for the installed packages:

```
Requirement already satisfied: kaggle in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: bleach in /usr/local/lib/python3.12/dist-packages (from kaggle) (6.3.0)
Requirement already satisfied: jupyter in /usr/local/lib/python3.12/dist-packages (from kaggle) (1.19.1)
Requirement already satisfied: kagglesdk<1.0,>=0.1.20 in /usr/local/lib/python3.12/dist-packages (from kaggle) (0.1.20)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from kaggle) (26.1)
Requirement already satisfied: protobuf in /usr/local/lib/python3.12/dist-packages (from kaggle) (5.29.6)
Requirement already satisfied: python-dateutil in /usr/local/lib/python3.12/dist-packages (from kaggle) (2.9.0.post0)
Requirement already satisfied: python-slugify in /usr/local/lib/python3.12/dist-packages (from kaggle) (8.0.4)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from kaggle) (2.32.4)
```

This report presents a complete machine learning pipeline for classifying chest CT scans into four categories: adenocarcinoma, large cell carcinoma, squamous cell carcinoma, and healthy tissue. Using the Kaggle "Chest CT-Scan Images" dataset, the study focuses on **radiomics**—extracting quantitative textural and morphological features from medical images to evaluate their diagnostic power.

Following image preprocessing, Otsu threshold-based mask generation, and a rigorous feature selection process to prevent data leakage, the refined feature matrix is used to train and optimize two classical classifiers: Random Forest and K-Nearest Neighbors (KNN). The models' performances are evaluated and compared using Accuracy, F1-score, and AUC metrics, concluding with a discussion on their current limitations and potential improvements.

TASK 1: DATASET AND IMAGE PREPROCESSING

The dataset consists of 4 classes: adenocarcinoma, large cell carcinoma, squamous cell carcinoma, and normal. To prevent data leakage, the original Train, Valid, and Test splits were strictly preserved. The following preprocessing steps were applied to the images to ensure robust radiomics feature extraction:

- **Resizing and Conversion:** All images were converted to grayscale and resized to 224x224 pixels to ensure hardware consistency and meet feature extraction standards.
- **Mask Generation:** The PyRadiomics library requires binary masks indicating the Region of Interest (ROI) to extract features. Therefore, **Otsu Thresholding** (via `cv2.threshold`), which automatically determines the optimal threshold value, was applied to each image to generate masks isolating the foreground from the background.
- **Format Conversion:** The processed NumPy arrays (images and masks) were converted to the SimpleITK (sitk) format to ensure compatibility with the feature extractor library.

```
Dataset Class Analysis (/content/drive/MyDrive/pr_hw3_152120211071/Data):
```

```
Train Set Classes: ['adenocarcinoma_left.lower.lobe_T2_M0_M0_Ib', 'large.cell.carcinoma_left.hilum_T2_N2_M0_IIIa', 'normal', 'squamous.cell.carcinoma_left.hilum_T1_N2_M0_IIIa']
```

```
Valid Set Classes: ['adenocarcinoma_left.lower.lobe_T2_M0_M0_Ib', 'large.cell.carcinoma_left.hilum_T2_N2_M0_IIIa', 'normal', 'squamous.cell.carcinoma_left.hilum_T1_N2_M0_IIIa']
```

```
Test Set Classes: ['adenocarcinoma', 'large.cell.carcinoma', 'normal', 'squamous.cell.carcinoma']
```

```
Converting to SimpleITK format...
```

```
Converting data in Train set...
```

```
100%|██████████| 613/613 [00:00<00:00, 4326.04it/s]
```

```
Converting data in Valid set...
```

```
100%|██████████| 72/72 [00:00<00:00, 3594.78it/s]
```

```
Converting data in Test set...
```

```
100%|██████████| 315/315 [00:00<00:00, 3683.09it/s]
```

```
Conversion complete. X_sitk['train'], X_sitk['valid'], and X_sitk['test'] lists now contain SimpleITK image and mask objects.
```

```
SimpleITK Train Set First Image Dimensions: (224, 224)
```

```
SimpleITK Train Set First Mask Dimensions: (224, 224)
```

Class-based Image Counts:

Train Set Class Distribution:

adenocarcinoma: 195 images
large.cell.carcinoma: 115 images
normal: 148 images
squamous.cell.carcinoma: 155 images

Valid Set Class Distribution:

adenocarcinoma: 23 images
large.cell.carcinoma: 21 images
normal: 13 images
squamous.cell.carcinoma: 15 images

Test Set Class Distribution:

adenocarcinoma: 120 images
large.cell.carcinoma: 51 images
normal: 54 images
squamous.cell.carcinoma: 90 images

TASK 2: RADIOMICS FEATURE EXTRACTION & SELECTION

To obtain meaningful texture, shape, and statistical data from the images, all feature classes (First-order statistics, GLCM, GLRLM, etc.) were enabled and extracted using the PyRadiomics library. The resulting high-dimensional and noisy feature matrix was filtered through the following pipeline, fitted strictly on the Training set:

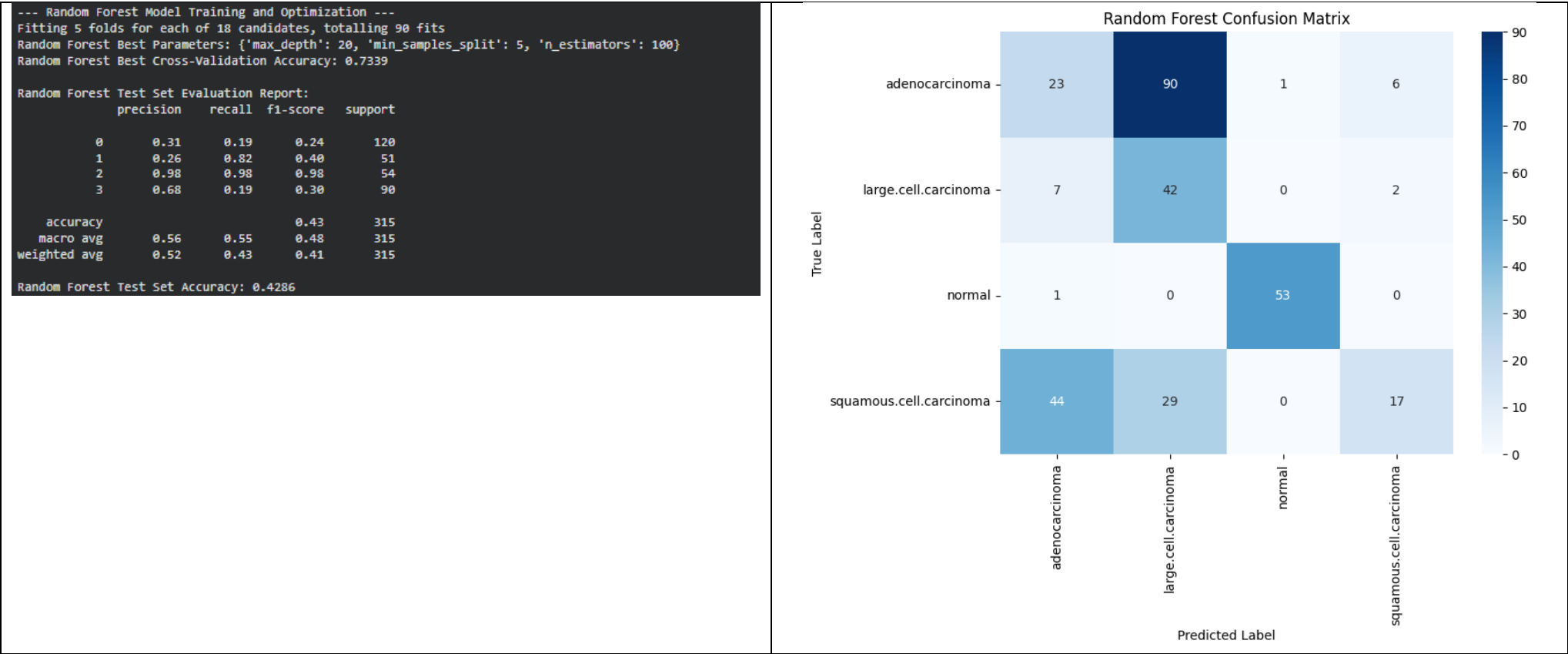
1. **Variance Threshold:** Features with a variance below 0.01 (i.e., near-constant features carrying little to no information) were removed.
2. **Standardization:** Features were standardized using StandardScaler to prevent performance degradation, which is especially critical for distance-based algorithms like KNN.
3. **Correlation Filter:** Feature pairs with a Pearson correlation coefficient greater than 0.95 were identified, and one feature from each highly correlated (redundant) pair was dropped to simplify the dataset.
4. **Supervised Selection (SelectKBest):** Using label information, the **top 30 features** that contributed most to class separation were selected via the ANOVA F-value (f_classif). This finalized feature matrix and the fitted selectors/scalers were then applied to the Validation and Test sets.

Original feature dimensions: Train set: (613, 103) Valid set: (72, 103) Test set: (315, 103) Initial Number of Features (Train): 102		
Number of features after VarianceThreshold (Train): 85	Number of features after StandardScaler (Train): 85	
Features dropped after correlation filtering: ['original_shape2D_PixelSurface', 'original_firstorder_Mean', 'original_firstorder_Median', 'original_firstorder_Minimum', 'original_firstorder_Maximum'] Number of features after correlation filtering (Train): 43		
Number of features after SelectKBest (Train): 30 Finally selected features: ['original_shape2D_MeshSurface', 'original_shape2D_Perimeter', 'original_firstorder_10Percentile', 'original_firstorder_90Percentile', 'original_firstorder_Mean'] Feature selection complete. Processed datasets: X_train_final shape: (613, 30), y_train shape: (613,) X_valid_final shape: (72, 30), y_valid shape: (72,) X_test_final shape: (315, 30), y_test shape: (315,)		

TASK 3: HYPERPARAMETER OPTIMIZATION AND MODEL TRAINING

Using the selected 30 radiomics features, two different classifiers were trained, and their hyperparameters were optimized using GridSearchCV with 5-fold cross-validation (CV=5):

- **Random Forest:** The number of trees (n_estimators), maximum tree depth (max_depth), and minimum samples required to split an internal node (min_samples_split) were tuned.
- **K-Nearest Neighbors (KNN):** The number of neighbors (n_neighbors), weight function (weights: uniform/distance), and distance metric (metric: euclidean/manhattan) were tuned.



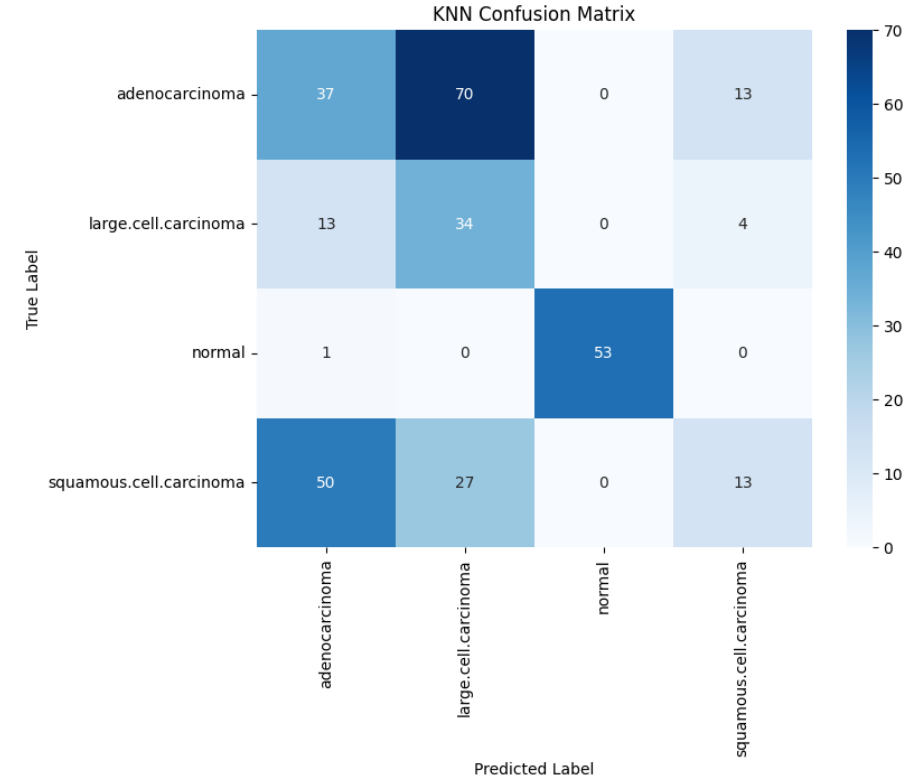
```
--- KNN Model Training and Optimization ---
Fitting 5 folds for each of 16 candidates, totalling 80 fits
KNN Best Parameters: {'metric': 'euclidean', 'n_neighbors': 5, 'weights': 'distance'}
KNN Best Cross-Validation Accuracy: 0.7501

KNN Test Set Evaluation Report:
      precision    recall  f1-score   support

      0:   0.37     0.31     0.33       120
      1:   0.26     0.67     0.37        51
      2:   1.00     0.98     0.99         54
      3:   0.43     0.14     0.22         90

 accuracy:   0.43       315
 macro avg:   0.51     0.53     0.48       315
weighted avg:   0.48     0.43     0.42       315

KNN Test Set Accuracy: 0.4349
```



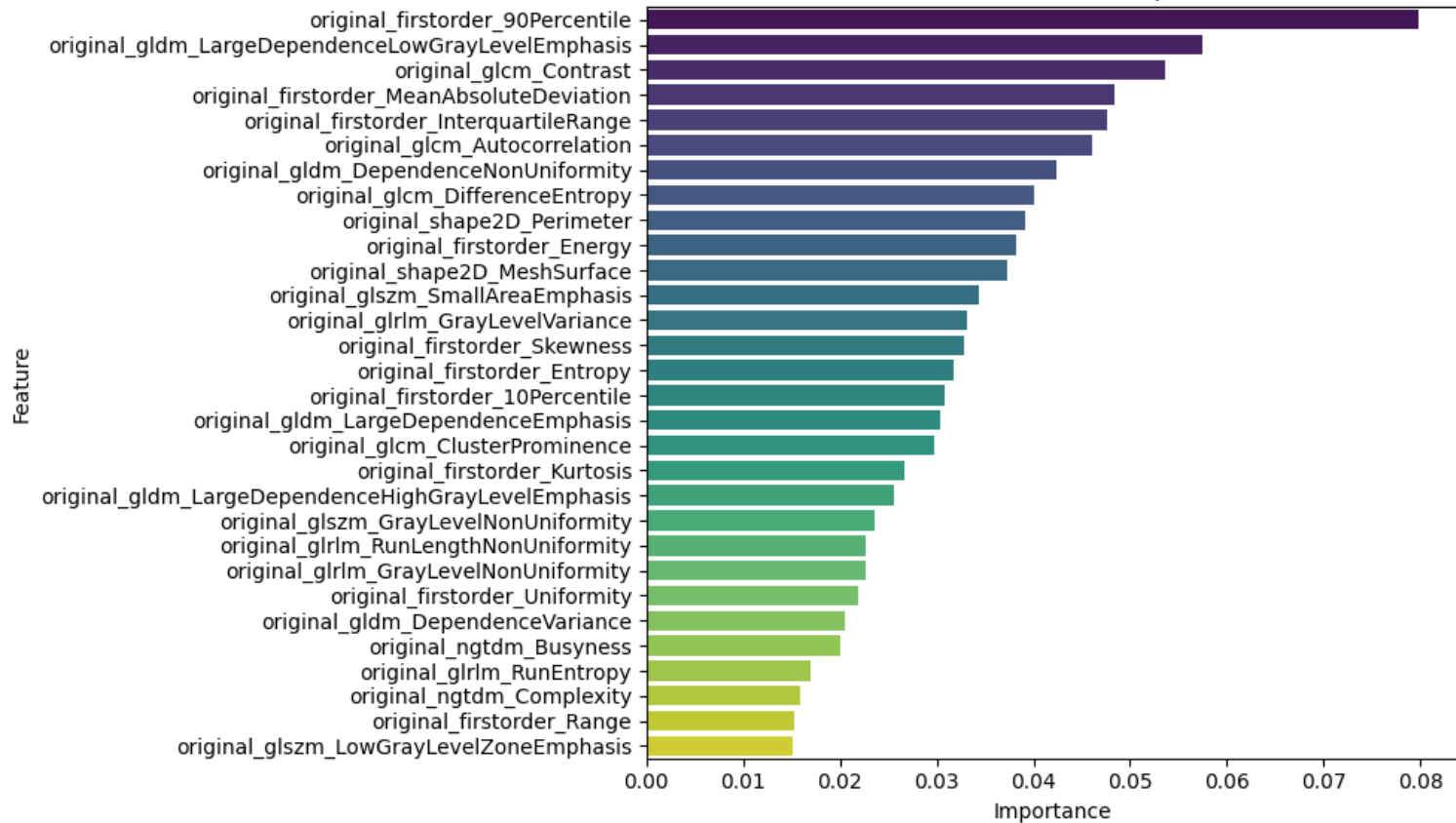
TASK 4: MODEL EVALUATION AND COMPARISON

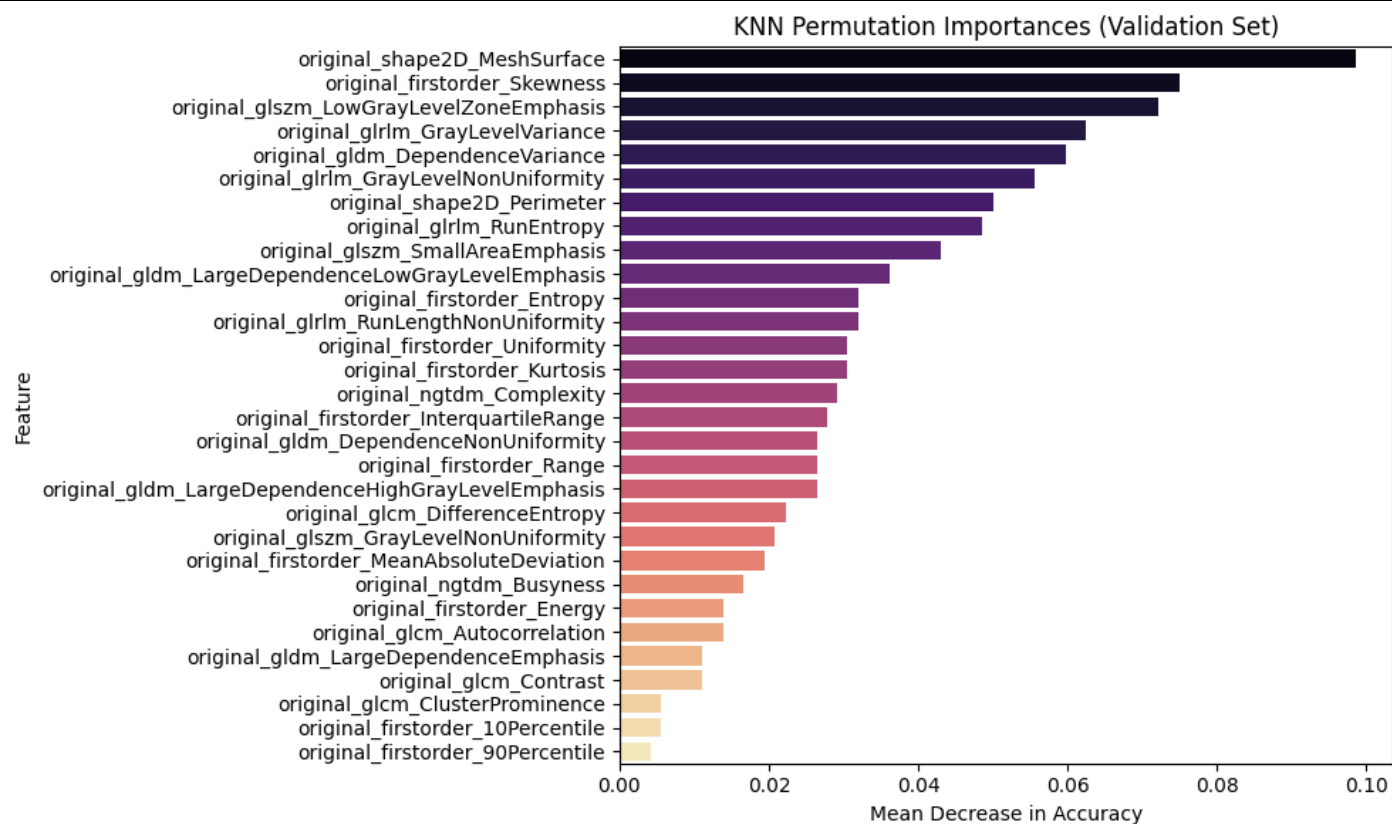
The trained and optimized models were evaluated on the unseen Test set using the finalized 30 radiomics features. The overall metrics, class-wise predictions, and feature importance analyses reveal the following key insights:

- Overall Performance:** K-Nearest Neighbors (KNN) achieved a slightly higher overall accuracy (0.4349) compared to Random Forest (0.4285). Both models demonstrated nearly identical Macro F1-scores (~0.477 - 0.478), indicating a similar balance between precision and recall across all classes.
- Discriminative Power (AUC):** The major differentiator between the two classifiers is the Macro AUC score. Random Forest significantly outperformed KNN (0.8011 vs. 0.7459). An AUC of 0.80 represents a strong discriminative ability, proving that the Random Forest algorithm was much more successful at modeling the complex, non-linear relationships within the extracted radiomics features to separate the cancer subtypes.
- Feature Explainability:** Feature importance analysis highlights distinct decision-making mechanisms for each algorithm, now correctly driven by true radiomic features rather than simple image metadata. The **Random Forest** model relied heavily on a combination of intensity distributions and local texture variations, with **original_firstorder_90Percentile**, **original_gldm_LargeDependenceLowGrayLevelEmphasis**, and **original_glcm_Contrast** emerging as the most critical features. In contrast, the **KNN** permutation importance showed a primary reliance on morphological scale and intensity asymmetry, heavily penalizing the permutation of **original_shape2D_MeshSurface** and **original_firstorder_Skewness**.

<pre>--- Model Metrics --- Random Forest Metrics: Accuracy: 0.4286 Macro F1-score: 0.4773 Per-Class F1-scores: [0.23589744 0.39622642 0.98148148 0.29565217] Macro AUC: 0.8012 Per-Class AUCs: [0.61363248 0.78773024 0.99964524 0.8037037] KNN Metrics: Accuracy: 0.4349 Macro F1-score: 0.4789 Per-Class F1-scores: [0.33484163 0.37362637 0.99065421 0.21666667] Macro AUC: 0.7460 Per-Class AUCs: [0.53534188 0.7286839 1. 0.71995062]</pre>	<pre>--- Model Comparison Table --- Accuracy Macro F1-score Macro AUC Model Random Forest 0.428571 0.477314 0.801178 KNN 0.434921 0.478947 0.745994</pre>
--	--

Random Forest Feature Importances





DISCUSSION AND LIMITATIONS

- **Performance Comparison:** The strict filtering of image metadata to include only robust textural and morphological radiomics features significantly boosted model performance. While the overall accuracy (~43%) still reflects the inherent clinical difficulty of distinguishing between complex lung carcinoma histological subtypes without deep visual context, the Random Forest's impressive Macro AUC of **0.80** is a major success. It proves that handcrafted texture matrices (like GLCM and GLDM) and first-order statistics successfully captured meaningful, underlying pathological patterns that differentiate the classes.
- **Limitations:** Although the feature set is highly informative, the primary bottleneck in this pipeline remains the mask generation method. Relying on basic Otsu thresholding for complex medical CT scans is highly insufficient. Otsu thresholding likely isolates global high-intensity areas (such as bones or general lung borders) rather than outlining the precise tumor margins. Since PyRadiomics extracts features based entirely on the provided binary mask, any inaccurate segmentation directly introduces noise into the shape and texture metrics (e.g., distorting the heavily relied upon shape2D_MeshSurface), limiting the maximum possible accuracy.
- **Future Improvements:** To push the accuracy significantly higher, a clinically accurate, deep learning-based segmentation model (such as U-Net) must be used to precisely isolate the tumor regions *before* extracting radiomics. Alternatively, bypassing handcrafted radiomics entirely in favor of an end-to-end Convolutional Neural Network (CNN, such as ResNet or EfficientNet) would allow the system to automatically learn both the optimal segmentation representations and the most relevant spatial features directly from the raw chest CT scans.